

Total Ionizing Dose Effects on Floating Gate Structures. Preliminary Results.

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Abstract—The design and characterization of an ionizing radiation sensor for radiotherapy applications based on floating gate structures are presented. The devices were fabricated in a commercial CMOS process, and the floating gate was extended over field oxide in order to improve the radiation sensitivity. Such devices are a useful tool for studying the total dose effects on floating gate structures. The sensor is provided with an injection electrode in order to manipulate the charge stored in the floating gate, allowing not only to set different charge conditions for a complete characterization, but also to restore the floating gate to a desired state after the charge was neutralized by radiation. The devices response to ionizing radiation was studied, showing that sensitivity increases as more charge is stored in the floating gate. The experimental results help to improve the modeling of total ionizing dose effects on floating gate structures and predict their degradation. Preliminary results show that the charge manipulation was successful in several samples, and also experimental results show good agreement with the expected theoretical sensitivity.

Index Terms—Total Ionizing Dose effects; Floating-Gate transistor; Dosimetry

I. INTRODUCTION

EXPOSURE to ionizing radiation affects electronic devices and circuits. There are several scenarios where electronic systems are exposed to ionizing radiation, such as Space Environments, Radiation Therapies in Health Institutions and Nuclear Facilities, among others. The main effect in MOS transistors is the variation of the threshold voltage due to positive charge trapped in the gate oxide [1]–[3]. Moreover, radiation-induced charge may also affect the charge stored in memory devices, with the consequent loss of information [4]. These effects compromise the electronic systems reliability, and that is why devices and circuits should be tested in order to assure their functionality in such hostile environments.

Besides reliability assurance, one can take advantage of the effects of ionizing radiation on electronic devices in order to measure radiation dose. The MOSFETs have been proposed as radiation dosimeters in 1970 [5] and much research has been

carried out since then. Nevertheless, there are still a series of issues that must be addressed.

In the last years, we have proposed the use of Field Oxide as a thick gate oxide (FOXFET), proving an improvement in the dosimeter sensitivity [6]. The FOXFET has shown to be appropriate for radiotherapy [7], even though the thick gate oxide also exhibited some instability in the trapped charge, manifested as fading. This dosimeter was also used to implement several differential measuring methods [8]–[11], but the differential method implies the use of different gate voltages applied each transistor, hence an electric connection with the dosimeter is needed, which is undesired in medical applications.

In the MOS devices family, the floating gate (FG) MOSFET is an alternative device to be used as a dosimeter, as shown in [12]–[16]. According to [13], [14], the FGMOSFET sensitivity can be improved extending the floating gate over the field oxide, in a similar way as the FOXFET, but also it presents the advantage that if the FG is charged, the electric field within the oxide can be maintained without external power supply [14], getting rid of the electrical connection during irradiation.

In this paper, we present the design and characterization of a floating gate MOSFET to be used as radiation sensor for radiotherapy applications. The device was designed in a standard 0.5 μm CMOS fabrication process in order to improve its sensitivity to radiation and also to make it possible to control the stored charge in the floating gate. The device was tested for charge manipulation in the FG, and also it was studied how the exposure to radiation neutralized the stored charge, affecting the drain current of the transistor. The device sensitivity to radiation was calculated for different floating gate stored charge, and results were compared to the expected theoretical sensitivity.

II. FLOATING GATE MOSFET DESIGN

The floating gate under test was designed and fabricated in a CMOS 0.5 μm commercial process. An schematic diagram of the device is shown in figure 1. The device is a p-channel MOSFET, where the polysilicon layer (P1L) acts as the floating gate itself and is extended over the Field Oxide to improve radiation sensitivity [6], [13], [14]. Both the

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gate oxide (gox) and the field oxide (fox) provide radiation generated charge [1] that will be collected in the floating gate (Q_{FG}), giving place to a shift in the floating gate potential (V_{FG}). To maximize the sensitivity of V_{FG} with radiation dose, the P1L over fox should be much larger than the P1L over gox, in order to counter the difference in oxide thickness, which are $t_{fox} \simeq 288 \text{ nm}$ and $t_{gox} \simeq 14 \text{ nm}$, respectively. A ratio $A_{fox}/A_{gox} = 100$ was chosen, and considering the width and length of the MOSFET to be $W = L = 20 \mu\text{m}$, the area for each P1L are $A_{fox} = 400 \mu\text{m} \times 100 \mu\text{m}$ and $A_{gox} = 20 \mu\text{m} \times 20 \mu\text{m}$.

An important aspect of the design process was to determine the layout for the injection of charge in the floating gate. Charging the floating gate occurs by Fowler-Nordheim (FN) tunneling from the injector electrode to the floating gate [17]. When a voltage is applied to the injector (V_{inj}) and all other terminals are short-circuited to ground, the floating gate (V_{FG}) potential due to the capacitive coupling of the conducting polysilicon strips is

$$V_{FG} = \frac{C_{inj} V_{inj} + Q_{FG}}{C_{inj} + C_{fox} + C_{gs} + C_{gd}}, \quad (1)$$

where C_{inj} is the capacitance between the injector and the floating gate, C_{fox} is the capacitance of the field oxide, C_{gs} and C_{gd} are the coupling capacitance from the floating gate to the source and drain respectively, and Q_{FG} is the charge accumulated in the floating gate. The gate oxide capacitance is divided between C_{gs} and C_{gd} so that $C_{gs} + C_{gd} = C_{gox}$, while the gate overlap capacitances ($C_{gs,ov}$ and $C_{gd,ov}$) are much smaller than the gate oxide capacitance and the field oxide capacitance ($C_{gs,ov}, C_{gd,ov} \ll C_{gox}, C_{fox}$). For injection to occur, V_{FG} must be different to V_{inj} , hence the voltage applied to the injector should drop mainly across the interpoly oxide. Considering $Q_{FG} \simeq 0$ in equation 1, this happens when $C_{fox} + C_{gs} + C_{gd} \simeq C_{fox} + C_{gox} \gg C_{inj}$.

The injection electrode was implemented with a second polysilicon layer (P2L) available, which lies above the P1L. Taking into account the capacitance ratio, the P2L area was designed as small as possible, considering the metal contact and the fabrication process specifications, resulting in $A_{inj} = 6 \mu\text{m} \times 6 \mu\text{m}$. The interpoly oxide thickness is $t_{inj} = 33 \text{ nm}$. No Control Gate was included over the floating gate as the parasitic capacitance of the control gate impoverishes radiation sensitivity [13].

A. Radiation Sensitivity

When the FG is exposed to ionizing radiation, electron-hole pairs are created as a result of the interaction between the ionizing particle and matter. For Silicon Oxide, the density of generated pairs per unit dose is $g_0 = 8.1 \times 10^{14} \text{ cm}^{-3} \text{ Gy}^{-1}$ [1]. Some of these pairs recombine immediately and only a fraction is able to transport through the oxide. The ratio of available pairs is called *fractional yield* (f_Y) and the amount is dependent on the electric field (E) applied to the oxide [18]. The total radiation generated charge is

$$\Delta Q_{rad} = q g_0 A t_{ox} f_Y(E) \Delta D \quad (2)$$

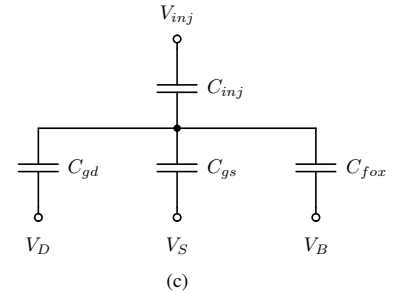
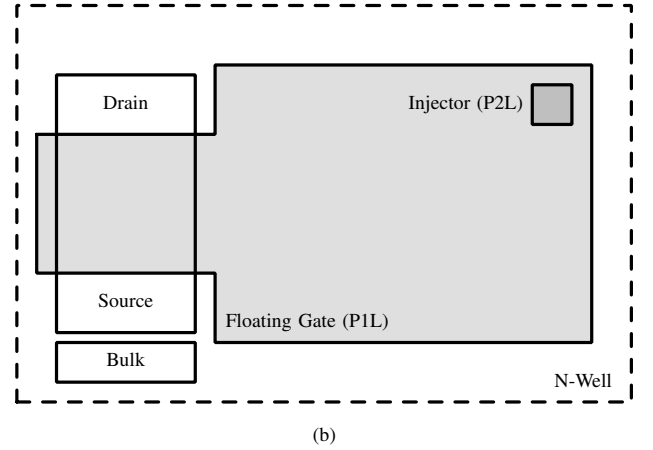
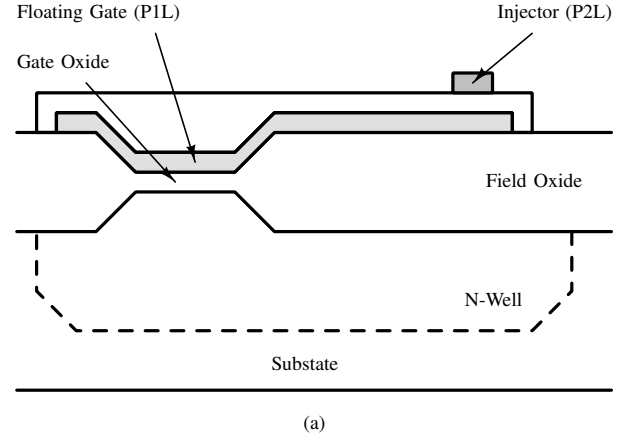


Fig. 1. Simplified diagram for the floating gate physical design [20]. (a) The cross-section shows how the different polysilicon layers overlap, and gives an idea of the different oxides involved in the device structure. (b) A simplified version of the layout to understand the different areas involved. For better understanding, metal, contacts and n/p diffusion layers were omitted. Bulk and Source are shorted from fabrication. (c) Capacitive equivalent model for the floating gate transistor.

where $q = 1.6 \times 10^{-19} \text{ C}$ is the elementary charge, the volume of the oxide is defined by its area (A) and thickness (t_{ox}) and ΔD is the radiation dose.

If the FG is filled with electrons, holes will be attracted towards the FG. Some of these holes may be captured within the oxide [19] and never reach the FG. The fraction of holes that effectively reach the FG is $(1 - f_T)$, where f_T is the fraction of trapped holes. Hence, the change in the stored

charge in the floating gate due to radiation is

$$\Delta Q_{FG} = (1 - f_T) \Delta Q_{rad}. \quad (3)$$

This change in the FG charge affects the FG potential. Considering that the injection electrode is biased only during the injection process, the change in V_{FG} can be written as

$$\frac{\partial V_{FG}}{\partial Q_{FG}} = \frac{1}{C_{fox} + C_{gox} + C_{inj}} = \frac{1}{C_{sum}}. \quad (4)$$

The drain current (I_D) is affected by the transistor transconductance

$$g_m = \frac{\partial I_D}{\partial V_{FG}} = 2\sqrt{k \cdot I_D}, \quad (5)$$

where $k = \frac{\mu}{2} \frac{C'_{gox}}{L} \frac{W}{L}$, μ is the holes mobility in the transistor channel, and C'_{gox} is the gate oxide capacitance per unit area.

Putting all together, the FG sensitivity to radiation can be written as

$$S = \frac{\partial I_D}{\partial D} = \frac{\partial I_D}{\partial V_{FG}} \frac{\partial V_{FG}}{\partial Q_{FG}} (1 - f_T) \frac{\partial Q_{rad}}{\partial D} \quad (6)$$

and replacing with equations (2), (4), (5):

$$S = \frac{2\sqrt{k \cdot I_D}}{C_{sum}} q g_0 A t_{ox} f_Y(E) (1 - f_T). \quad (7)$$

The radiation induced charge is generated in every oxide in the FG structure: the field oxide, the gate oxide and the interpoly oxide between the injection electrode and the floating gate. The total charge generated is then the sum of all these contributions, and as each oxide has a different electric field applied, the fraction of charge that reach the floating gate is different. The total charge collected in the FG is then

$$\Delta Q_{FG} = q g_0 \sum_i (1 - f_{T,i}) A_i t_{ox,i} f_{Y,i}(E_i) \Delta D. \quad (8)$$

Taking all contributions into consideration, a more general expression for the radiation sensitivity is

$$S = \frac{2\sqrt{k \cdot I_D}}{C_{sum}} q g_0 \sum_i A_i t_{ox,i} f_{Y,i}(E_i) (1 - f_{T,i}). \quad (9)$$

Nevertheless, when each contribution for eq. (8) is examined, it is found that the contribution of the field oxide is predominant, as its volume is much larger than the one of the gate oxide or the injection oxide: $A_{fox} \cdot t_{fox} = 2057 \times A_{gox} \cdot t_{gox} = 9697 \times A_{inj} \cdot t_{inj}$. Hence, eq. (9) can be reduced back to eq. (7). Moreover, the capacitance C_{sum} can be expressed in terms of the field oxide geometric parameters as

$$C_{sum} = \epsilon_{ox} \left(1 + \frac{C_{gox}}{C_{fox}} + \frac{C_{inj}}{C_{fox}} \right) \frac{A_{fox}}{t_{fox}} = \epsilon_{ox} \frac{1}{\alpha} \frac{A_{fox}}{t_{fox}}, \quad (10)$$

where $\alpha = 0.824$ in our device. Replacing eq. (10) in eq. (7) we obtain

$$S = 2\sqrt{k \cdot I_D} \alpha \frac{t_{fox}^2}{\epsilon_{ox}} q g_0 f_Y(E) (1 - f_T). \quad (11)$$

Equation 11 shows the device sensitivity dependence with the square of the field oxide thickness. The α factor justifies

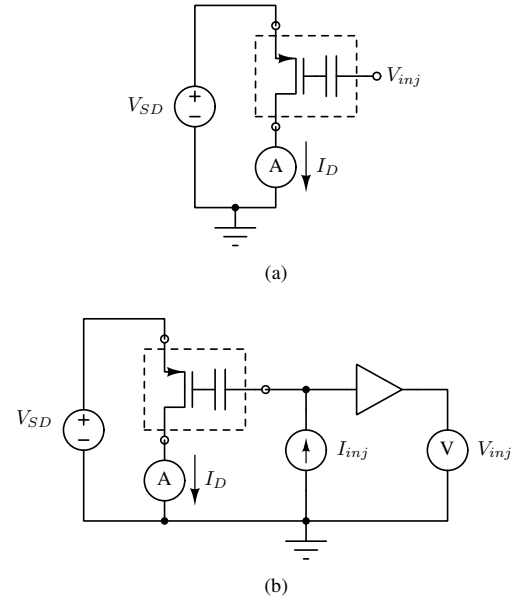


Fig. 2. Schematic circuits for (a) the current-voltage measurement for the FG MOSFET, and (b) the charge injection process [20].

not only a design with a floating gate extended over field oxide for a high sensitive device, but also the high area ratio between the floating gate over field oxide and the floating gate over gate oxide.

III. RESULTS

A. Experimental measurements of the injection process

The floating gate was designed and fabricated. As this is the first characterization of the device, the fabricated dies were not wire-bonded nor encapsulated. The electrical characterization was performed in a probe station for four different samples.

Figure 2 shows the schematic circuits for both the current-voltage (IV) measurement for the FG MOSFET (fig. 2.a) and the charge injection process in the floating gate (fig. 2.b). For the IV measurement, two Keithley 617 electrometers were used, one as a voltage source (V_{SD}) and the other one as an ammeter (I_D). The voltage source was swept while measuring the drain current (I_D). The initial charge condition in the FG was different for each sample. Some of them were discharged, and no current flowed since no channel was formed beneath the gate of the device, and another batch was already charged from fab, so a typical transistor output characteristic (I_D vs V_{SD}) was measured.

A key aspect of the FGMOSFET is that the FG itself can be charged or discharged in order to change its operation state. As the samples are p-channel MOSFETs, the FG should be charged with electrons to induce a conduction channel underneath the interface. For charge injection, a Keithley 220 programmable current source was connected to the injection electrode while a constant $V_{SD} = 100$ mV was applied to the MOSFET, and the I_D was monitored. A buffered output from the current source allowed to monitor the injection voltage (V_{inj}) with one of the Keithley 617 as voltmeter.

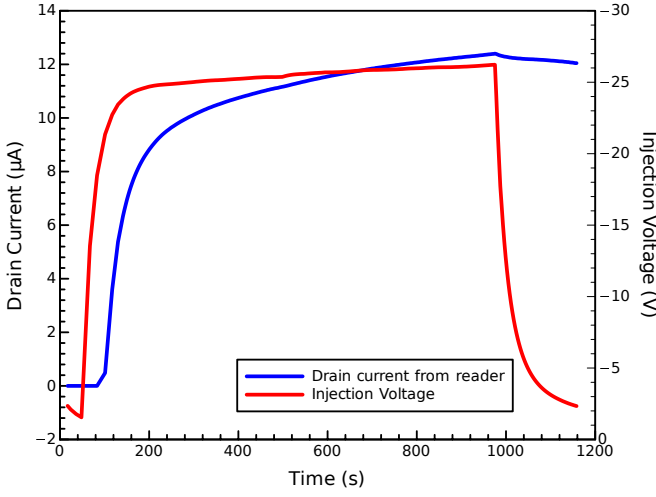


Fig. 3. Change in the drain current (blue line) due to charge injection in the floating gate for one of the samples. The voltage applied to the injection electrode is shown (red line). Injection begins approximately at $t = 120$ s when $V_{inj} = -22$ V.

In order to charge the samples with electrons, a negative current (I_{inj}) was applied, whereas when discharging the samples a positive I_{inj} was applied instead. In both cases, the injection process was similar. V_{inj} increases in absolute value as the injection node charges in time until it reached approximately $V_{inj} = -22$ V when charging and $V_{inj} = 18$ V when discharging. At these voltages, the injection starts, and the drain current will change in time. An example of the injection process is shown in figure 3 for one of the initially discharged samples.

B. Radiation sensitivity measurements

To test the device sensitivity to radiation and its dependence with the drain current the following experiment was performed. First, the four samples under test were submitted to an injection process in order to set the initial drain current to approximately $300 \mu\text{A}$, and the IV curve was registered for each sample. Afterwards, the samples were irradiated up to around approximately 1 Gy dose, and the IV curve was measured again to track the shift in the output characteristic. Then the experiment was repeated extracting charge from the FG to shift the drain current of the four samples to approximately $200 \mu\text{A}$, and 1 Gy was delivered again and the shift was tracked measuring the IV curve. The third time the experiment was carried out, the drain current was lowered to approximately $100 \mu\text{A}$, and another 1 Gy was delivered.

The irradiation of the samples was performed with a radiotherapy unit, the INVAP Teradi 800, with a ^{60}Co radiation source. The samples were laid over the stretcher of the unit, with no electrical connection, at 80 cm from the source, where the dose rate was estimated to be 0.4 Gy/min . The irradiation time was set to 2.35 min, delivering a total dose of 0.94 Gy in each irradiation.

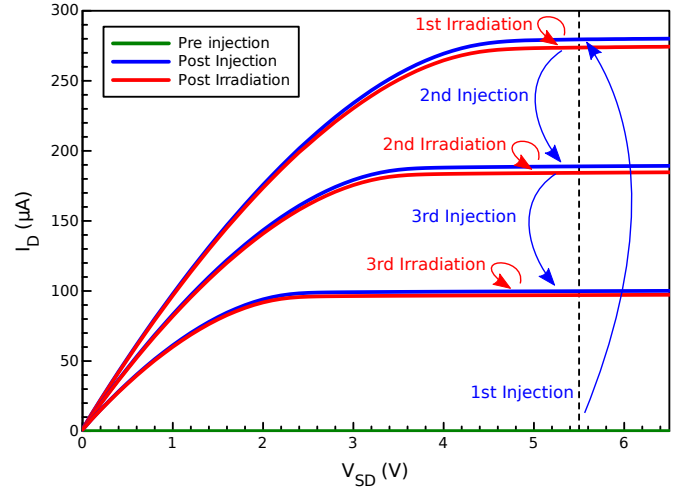


Fig. 4. Current-Voltage curves of the floating gate after each injection process and after each irradiation session. The dashed line shows the current taken to calculate the sensitivity.

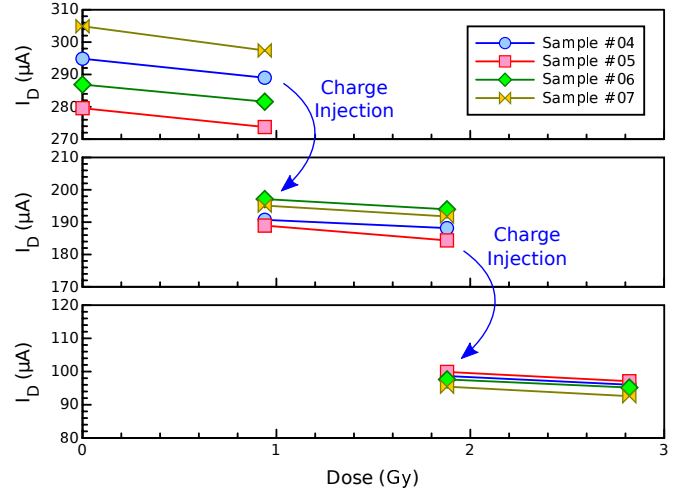


Fig. 5. Evolution of the drain current of all samples with absorbed dose. Between irradiation sessions, charge was injected in the floating gate to change the operation condition of the device.

The IV curves for one of the samples after each stage of the experimental procedure are shown in fig. 4. If we track the drain current at $V_{SD} = 5.5$ V, we obtain the results shown in fig. 5 where all tested samples are shown. From the variation of the drain current in each radiation session, the sensitivity was calculated as $S = |\Delta I_D / \Delta D|$ where $\Delta D = 0.94 \text{ Gy}$, and the result is plotted against the average drain current in fig. 6.

IV. DISCUSSION AND CONCLUSION

In this paper, the design and first characterization of a floating gate transistor for dosimetry applications was presented. The device was fabricated in a CMOS $0.5 \mu\text{m}$ commercial process. The designed device is a MOS transistor where the isolated gate electrode is extended over field oxide in order to improve total ionizing dose sensitivity. An injection electrode

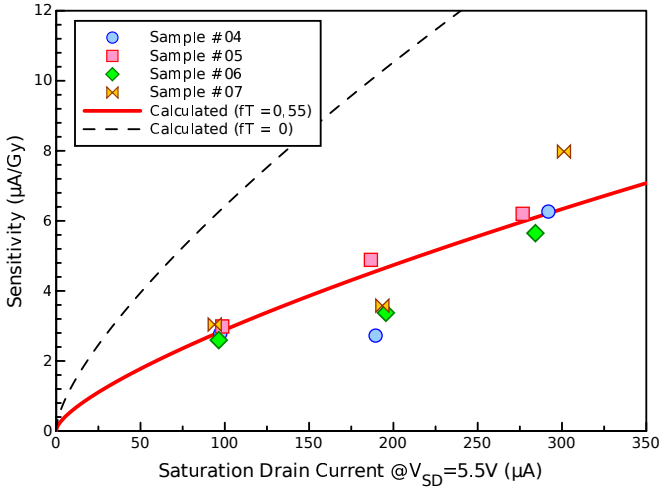


Fig. 6. Estimated sensitivity of the device under test as a function of the drain current. Symbols show measurements and lines show the calculated response. The dashed black line considers that all generated charge reaches the floating gate, whereas the red solid line takes into consideration that 55% of the generated charge is trapped in the field oxide and never reaches the floating gate.

was implemented using a second polysilicon layer over the gate layer in order to manipulate the charge in the floating gate.

A batch of four samples was electrically characterized. It was found that the initial charge condition was different between samples. Nevertheless, the injection process was successfully performed over all the samples, allowing to set the drain current of the devices under test to $291 \mu\text{A} \pm 3.7\%$ before the first irradiation test; $193 \mu\text{A} \pm 2.0\%$ after the first irradiation; and $98 \mu\text{A} \pm 1.9\%$ before the last irradiation. The injection process was repeatable, and it was accurately modeled by Fowler-Nordheim currents as shown in [20].

The irradiation test showed that the charge stored in the floating gate is neutralized by the radiation induced charge, as the drain current in the transistors lowers after exposed to radiation. The sensitivity was calculated from measurements and results were compared with the theoretical sensitivity in equation (11). Equation (11) considers that the charge generated in the field oxide is the main responsible for the shift in the drain current. To validate this statement, the radiation induced charge in every oxide (field, gate and interpoly) was calculated. The results shown in fig. 7 demonstrate that this consideration was reasonable, and that all other charge contributions can be neglected. That being the case, the sensitivity expressed in equation (11) worth some analysis.

First of all, the sensitivity is proportional to the square root of the drain current. The higher the drain current, the more sensitive the device should be. In the span of our experiments, this was confirmed as sensitivity for $I_D = 96.6 \mu\text{A} \pm 1.7\%$ was $2.8 \mu\text{A Gy}^{-1} \pm 6\%$, and increased when the drain current was $I_D = 191 \mu\text{A} \pm 2.1\%$ to $3.6 \mu\text{A Gy}^{-1} \pm 25\%$ and to $6.5 \mu\text{A Gy}^{-1} \pm 15\%$ when $I_D = 288 \mu\text{A} \pm 3.6\%$. Secondly, the sensitivity is proportional to the square of the field oxide

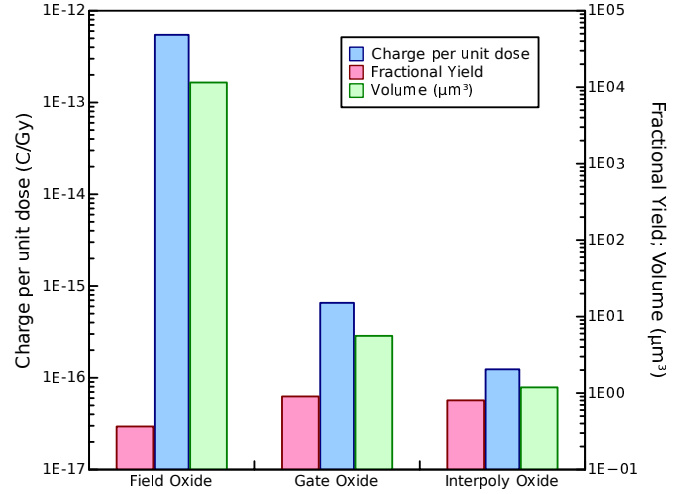


Fig. 7. Calculated charge per unit dose in each oxide volume. It can be seen that most of the charge is generated in the field oxide due to its greatest volume. The difference in fractional yield, which is larger for the gate oxide, does not compensate for the difference in volume.

thickness, as expected from previous results for thick oxide MOSFETs [21]. It is important to notice that to take advantage of the field oxide sensitivity, special considerations should be taken in the design of the floating gate device, i.e. the ratio between the area of the field oxide and the gate oxide must be large. This impacts directly in the factor α . In our case, considering that $A_{fox}/A_{gox} = 100$, $\alpha = 0.824$. To improve sensitivity, a larger ratio should be chosen, whereas if the design is limited in area, sensitivity will be lost.

Finally, in order for equation (11) to fit our experimental results, not all generated charge should reach the floating gate. It was found that 55% of the generated charge should be trapped in the field oxide before reaching the floating gate. This impacts poorly in the response of the device, as if all generated charge reached the floating gate, the sensitivity would be double. The black dashed curve in fig. 6 shows the best case, where no charge is trapped within the oxide, which is much higher than the red curve that best fits our results, considering $f_T = 0.55$.

In conclusion, preliminary results on the response of floating gate devices to total ionizing dose were presented, which fit adequately the theoretical sensitivity considering that the main contribution comes from the field oxide. These first results show large variation, specially in the first two irradiation sessions, as noted by the standard deviation in the previous paragraphs. For the third session, the measurement protocol was improved, and the results were more precise. Future work will be aimed to get a better set of measurements in order to confirm these preliminary results and also to be able to refine the modeling of the response of the devices and complete the calibration of the sensors.

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